

HF Controls Ltd

100 Albert Drive, Glasgow, G41 2SJ

www.hf-group.co.uk

Call 0800 783 9480

Glasgow

Edinburgh

Manchester

Belfast

Aberdeen

PROJECT: RSGT, LIVERPOOL

PROJECT LOCATION: SWITCHROOM 10

PROJECT REFERENCE NUMBER: 284595

CLIENT: PEEL PORTS

CLIENT ADDRESS: MARITIME CENTRE PORT OF LIVERPOOL

CLIENT ORDER NUMBER: ????

CONTROL SYSTEM DETAILS & SPECIFICATION

SYSTEM TYPE: SAFE AREA

ENCLOSURE MATERIAL: MILD STEEL

RATED VOLTAGE: 230V AC

RATED FREQUENCY: 50 Hz

RATED AMPERE:	16A
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CONTROL VOLTAGE: 24V DC

DEGREE OF PROTECTION: IP 55

PAINT FINISH: POWDER COATED RAL7035

YEAR OF MANUFACTURE:	2022
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REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BALDOCK	
00	24/06/24	AS FITTED			DRAWN	D.BALDOCK	
01	07/01/24	SPARE INPUTS UTILISED DUST PLANT 110V SUPPLIES ADDED			CHECKED	M.WRIGHT	
					DATE	11/05/2023	
02			INSTALL	RSGT LIVERPOOL SWITCHROOM 10	DRG. No.:		
03					284595_3111		
04							
05			TITLE	CP10 NON-I.S. PLC PANEL WIRING SCHEMATICS	DRAWING REVISION :	PAGE No.:	
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<table><tr><td>CURRENT CIRCUIT</td><td>CONDUCTOR</td><td>WIRING COLOUR</td><td>MINIMUM CROSS-SECTION</td></tr><tr><td>3-PHASE POWER</td><td>L1</td><td>BROWN</td><td>2.5mm2</td></tr><tr><td>3-PHASE POWER</td><td>L2</td><td>BLACK</td><td>2.5mm2</td></tr><tr><td>3-PHASE POWER</td><td>L3</td><td>GREY</td><td>2.5mm2</td></tr><tr><td>3-PHASE NEUTRAL</td><td>N</td><td>LIGHT BLUE</td><td>2.5mm2</td></tr><tr><td>1-PHASE POWER</td><td>L1</td><td>BROWN</td><td>1.5mm2</td></tr><tr><td>1-PHASE POWER</td><td>N</td><td>LIGHT BLUE</td><td>1.5mm2</td></tr><tr><td>24VDC</td><td>+</td><td>DARK BLUE</td><td>0.75mm2</td></tr><tr><td>24VDC</td><td>-</td><td>DARK BLUE</td><td>0.75mm2</td></tr><tr><td>MEASURING</td><td>+/-</td><td>GREY</td><td>0.5mm2</td></tr><tr><td>EXTERNAL VOLTAGE</td><td></td><td>ORANGE</td><td>1.5mm2</td></tr><tr><td>PROTECTIVE EARTH CONDUCTOR</td><td>PE</td><td>GREEN/YELLOW</td><td>1.5mm2</td></tr></table>						CURRENT CIRCUIT	CONDUCTOR	WIRING COLOUR	MINIMUM CROSS-SECTION	3-PHASE POWER	L1	BROWN	2.5mm2	3-PHASE POWER	L2	BLACK	2.5mm2	3-PHASE POWER	L3	GREY	2.5mm2	3-PHASE NEUTRAL	N	LIGHT BLUE	2.5mm2	1-PHASE POWER	L1	BROWN	1.5mm2	1-PHASE POWER	N	LIGHT BLUE	1.5mm2	24VDC	+	DARK BLUE	0.75mm2	24VDC	-	DARK BLUE	0.75mm2	MEASURING	+/-	GREY	0.5mm2	EXTERNAL VOLTAGE		ORANGE	1.5mm2	PROTECTIVE EARTH CONDUCTOR	PE	GREEN/YELLOW	1.5mm2
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EXTERNAL VOLTAGE		ORANGE	1.5mm2																																																		
PROTECTIVE EARTH CONDUCTOR	PE	GREEN/YELLOW	1.5mm2																																																		

MEASURING LINES:

- ☒ SINGLE CONDUCTOR
☐ TWISTED PAIRS
☒ ANALOGUE SIGNAL IN SHIELDED PAIRS

TYPE OF CONNECTION:

- ☒ WIRE END BOOTLACE CRIMP
☐ WIRE END PIN CRIMP
☐ FORK TYPE CRIMP
☐ ISOLATED
☐ NO CRIMP

WIRE MARKING:

- ☐ NONE
☐ WIRE DESIGNATION, SEMI OPEN
☐ WIRE TARGET DESTINATION
☒ CURRENT PATH DESIGNATION

SPECIFICS:

- ☐ HALOGEN FREE
☒ IP20 FOR INFEED TERMINALS
☒ BUS-BAR PE

CONSTRUCTION NOTES:

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1.  TERMINALS:
    POWER      WDU4
    INSTRUMENT  WDU2.5
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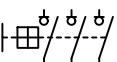


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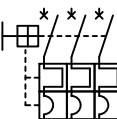
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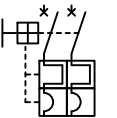
CODE OF ELECTRICAL SYMBOLS



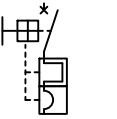
Q = MAIN ISOLATING SWITCH



QFM = ISOLATING CIRCUIT BREAKER
(3 POLE)



QFM = ISOLATING CIRCUIT BREAKER
(2 POLE)



QFM = ISOLATING CIRCUIT BREAKER
(1 POLE)



CONTACT (POWER)
NORMALLY OPENED



CONTACT-THERMOSTAT
NORMALLY CLOSED



FU = OVERLOAD PROTECTIVE DEVICE
(3 POLE)



SF = SELECTOR SWITCH
2 POSITION / STAY PUT



SF = PUSH BUTTON / SPRING RETURN
NORMALLY OPEN



SF = PUSH BUTTON / SPRING RETURN
NORMALLY CLOSED



Kx = CONTROL COIL



PLC INPUT



PLC OUTPUT



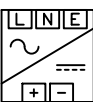
INDICATION LED



HEATER

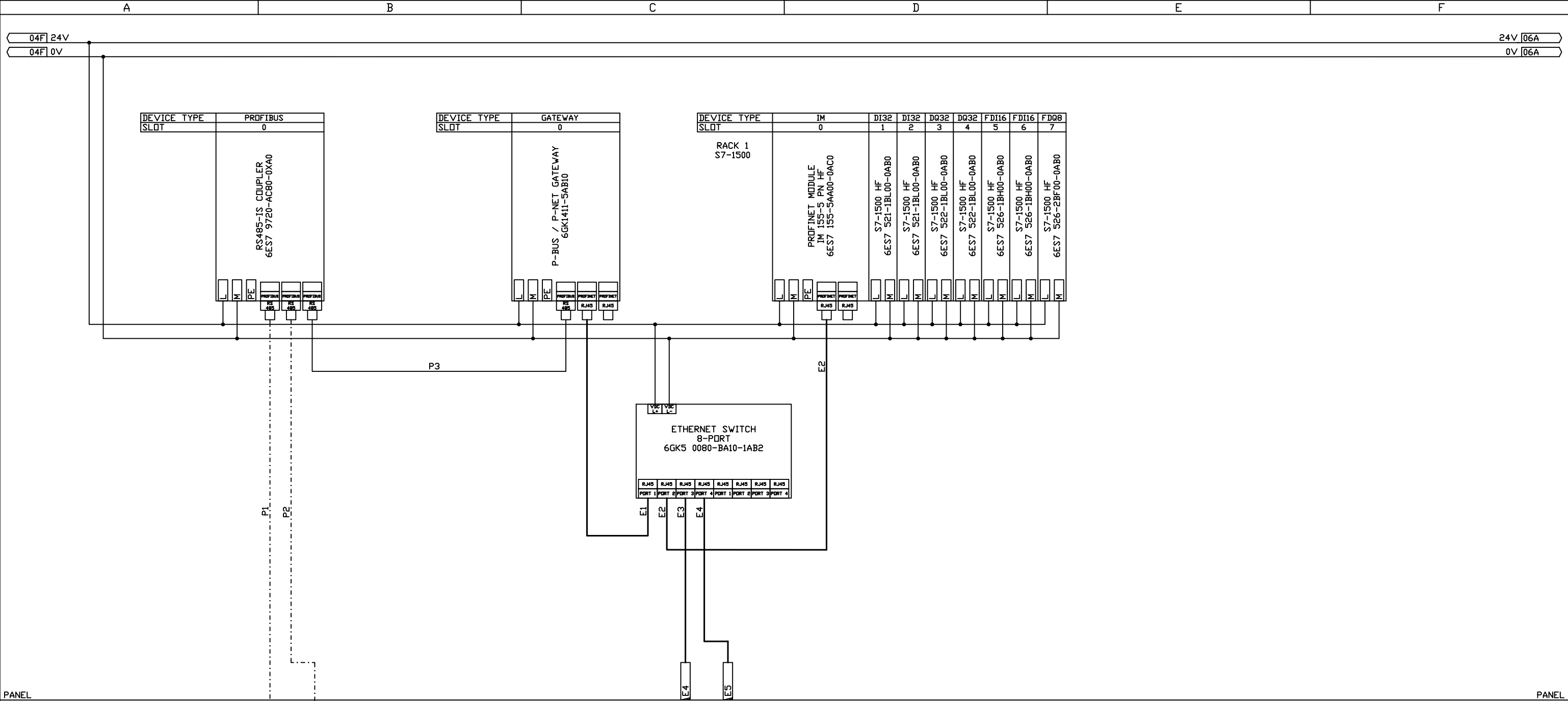


RESISTOR



PSU = AC TO DC POWER CONVERSION

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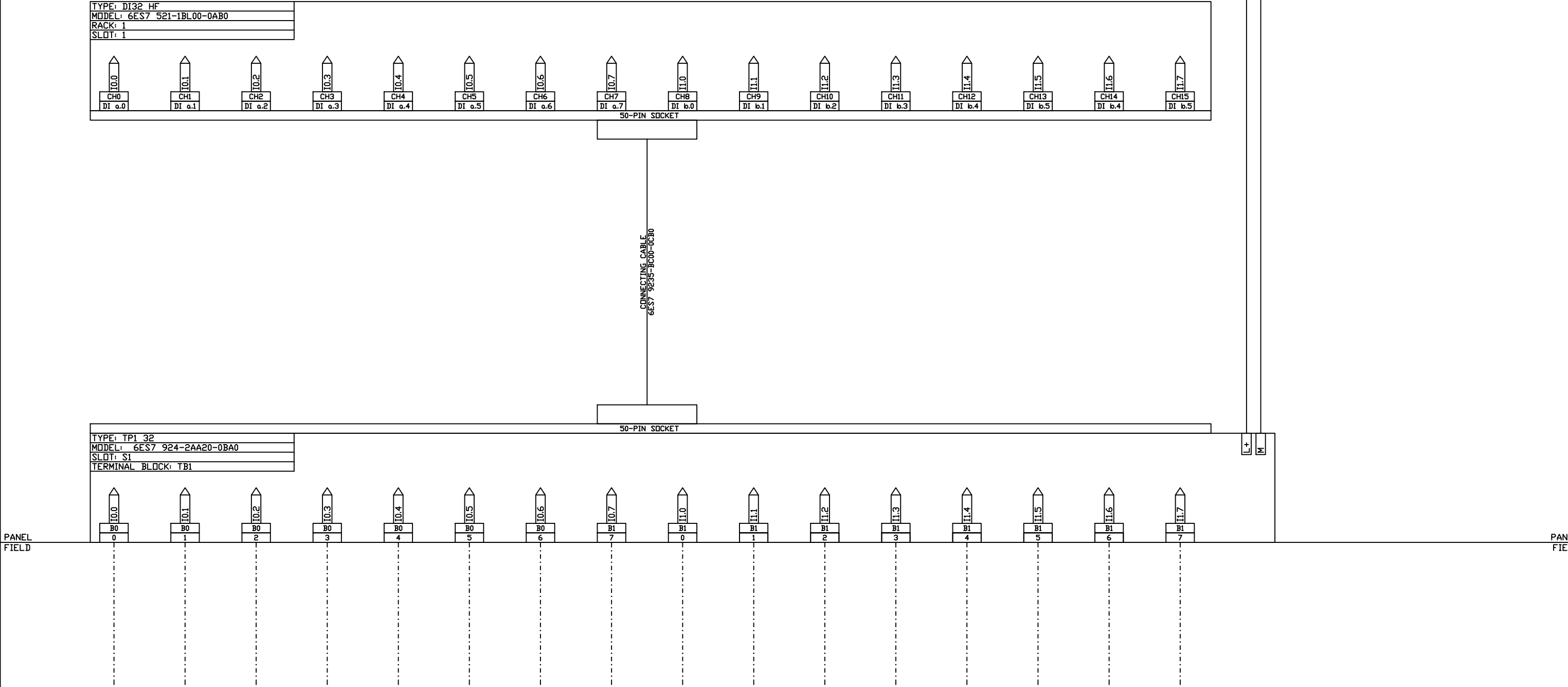
		PROFIBUS IN	PROFIBUS OUT			ETHERNET COMMS FROM MCC2A	ETHERNET COMMS FROM MCC2C										
 HF Controls Ltd 100 Albert Drive, Glasgow, G41 2SJ Call 0800 783 9480 www.hf-group.co.uk  Glasgow Edinburgh Manchester Belfast Aberdeen								REV	DATE	DETAIL	CLIENT	PEEL PORTS	ENGINEERED	D.BALDOCK			
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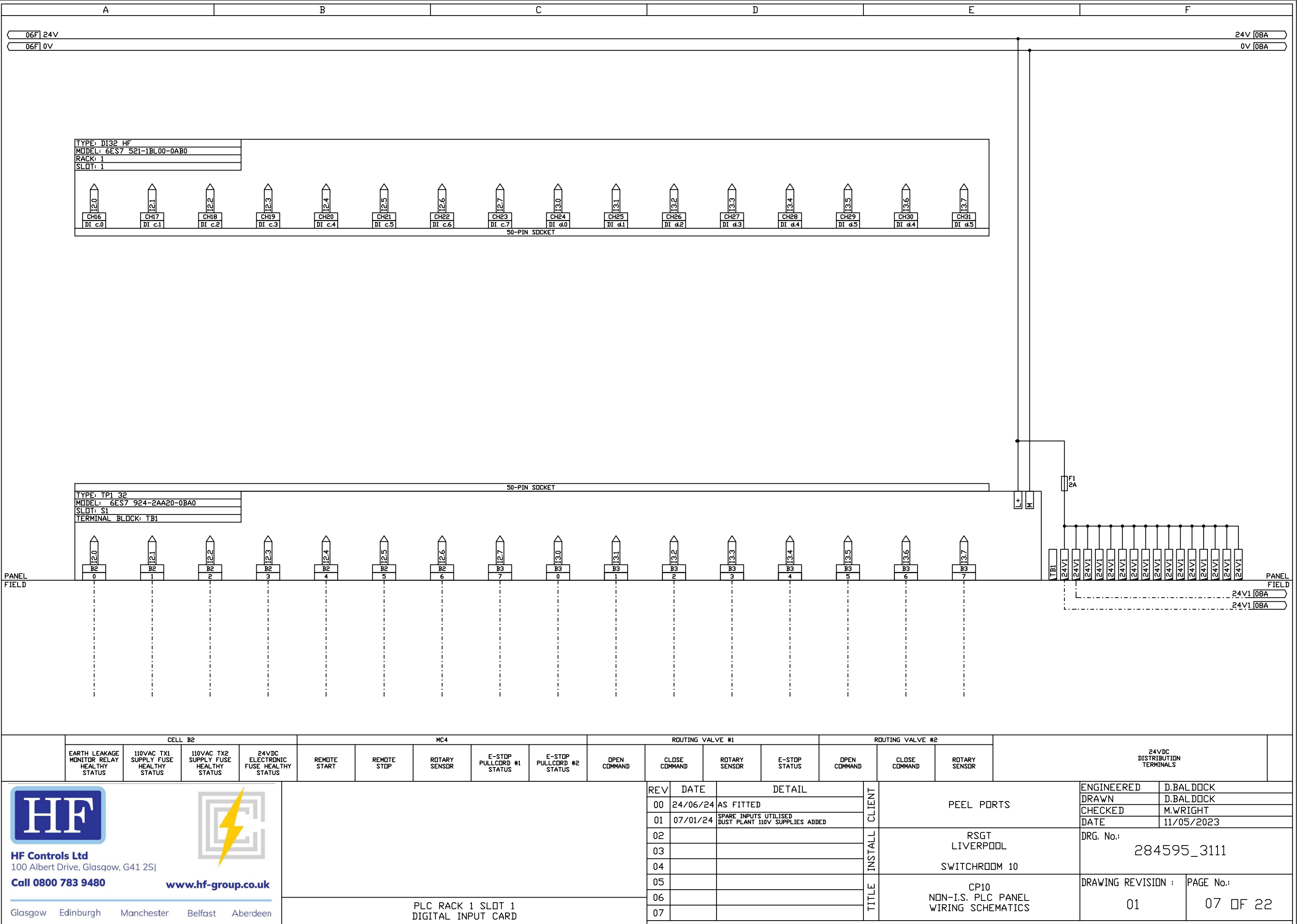


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TYPE: TP1 32

MODEL: 6ES7 924-2AA20-0BA0

SLOT: S1

TERMINAL BLOCK: TB1

12.0

B2

0

12.1

B2

1

12.2

B2

2

12.3

B2

3

12.4

B2

4

12.5

B2

5

12.6

B2

6

12.7

B3

7

13.0

B3

0

13.1

B3

1

13.2

B3

2

13.3

B3

3

13.4

B3

4

13.5

B3

5

13.6

B3

6

13.7

B3

7

50-PIN SOCKET

+

N

F1

2A

24V1

08A

24V1

08A

PANEL FIELD

PANEL FIELD

HF

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PLC RACK 1 SLOT 1

DIGITAL INPUT CARD

REV	DATE	DETAIL
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03		
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CLIENT

INSTALL

TITLE

PEEL PORTS

RSGT LIVERPOOL

SWITCHROOM 10

CP10

NON-I.S. PLC PANEL

WIRING SCHEMATICS

ENGINEERED

DRAWN

CHECKED

DATE

DRG. No.:

DRAWING REVISION :

PAGE No.:

D.BALDOCK

D.BALDOCK

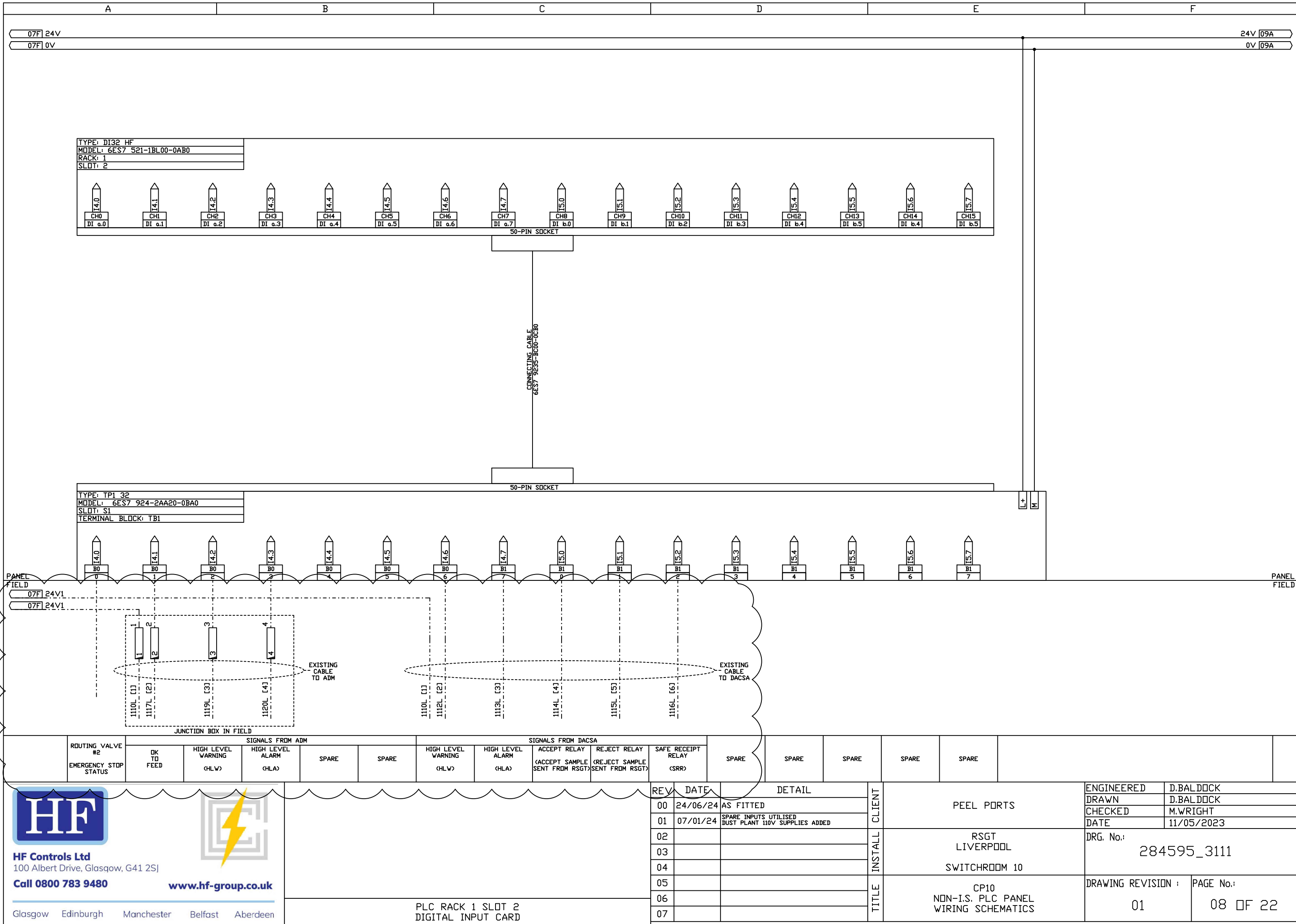
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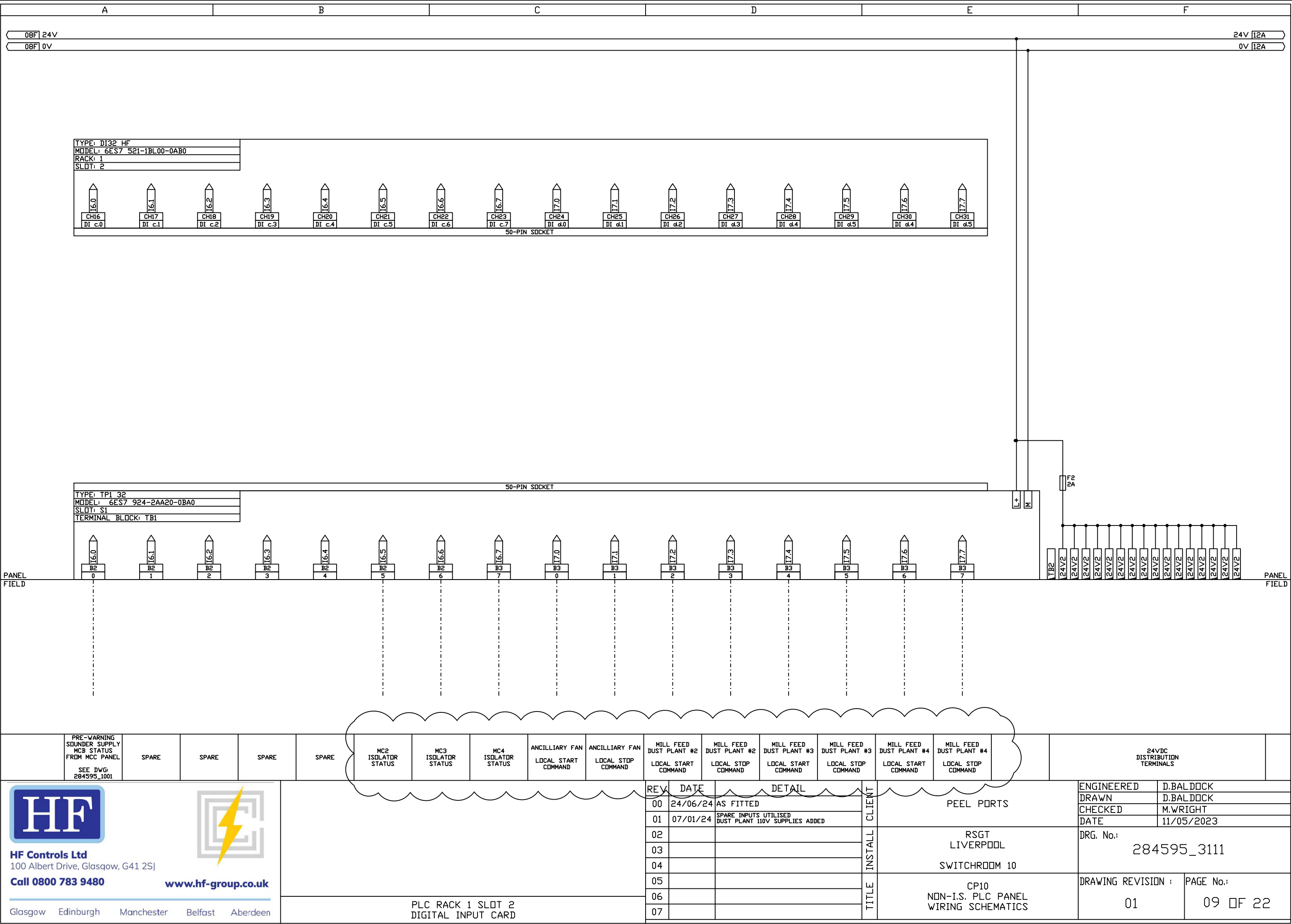
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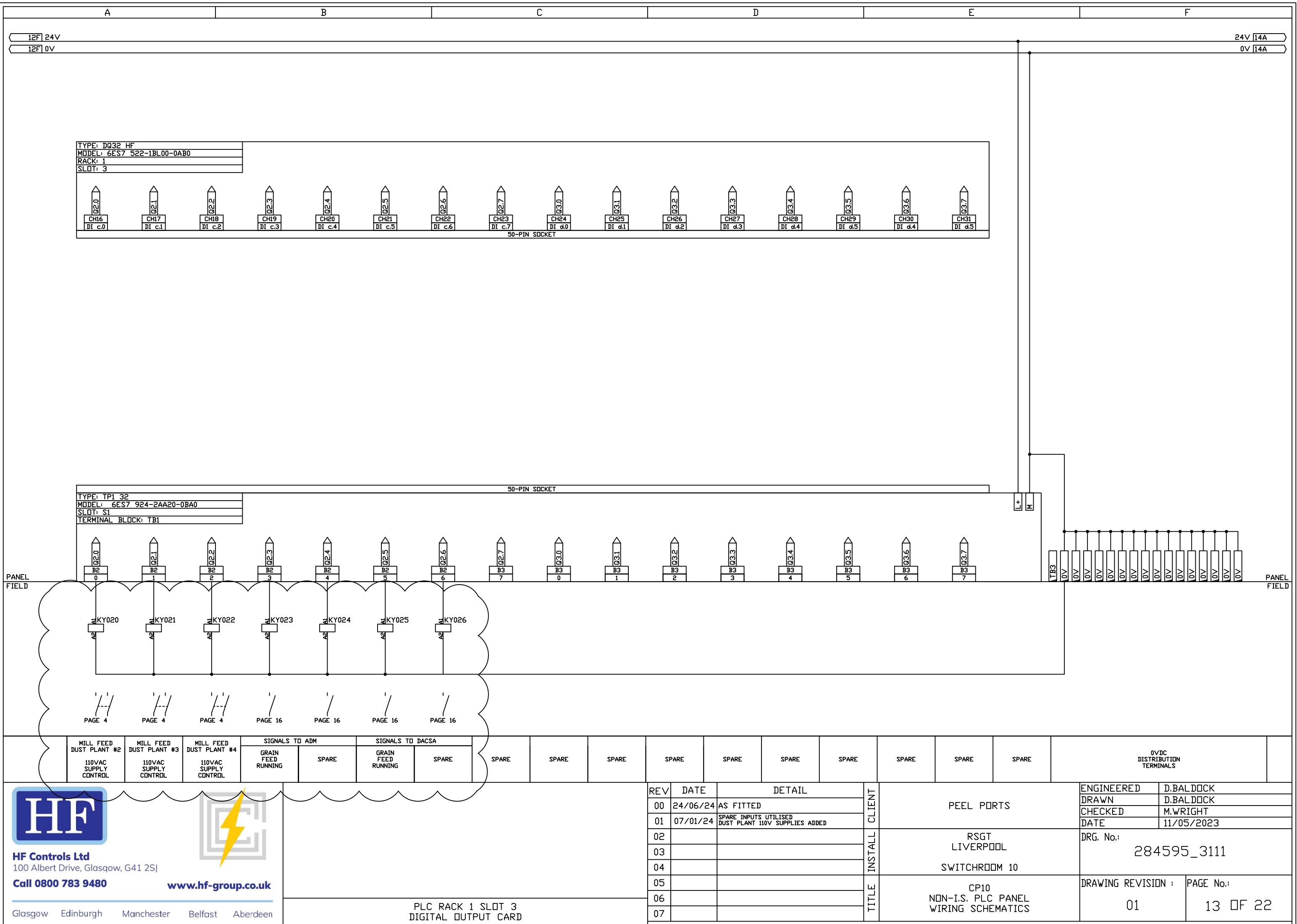
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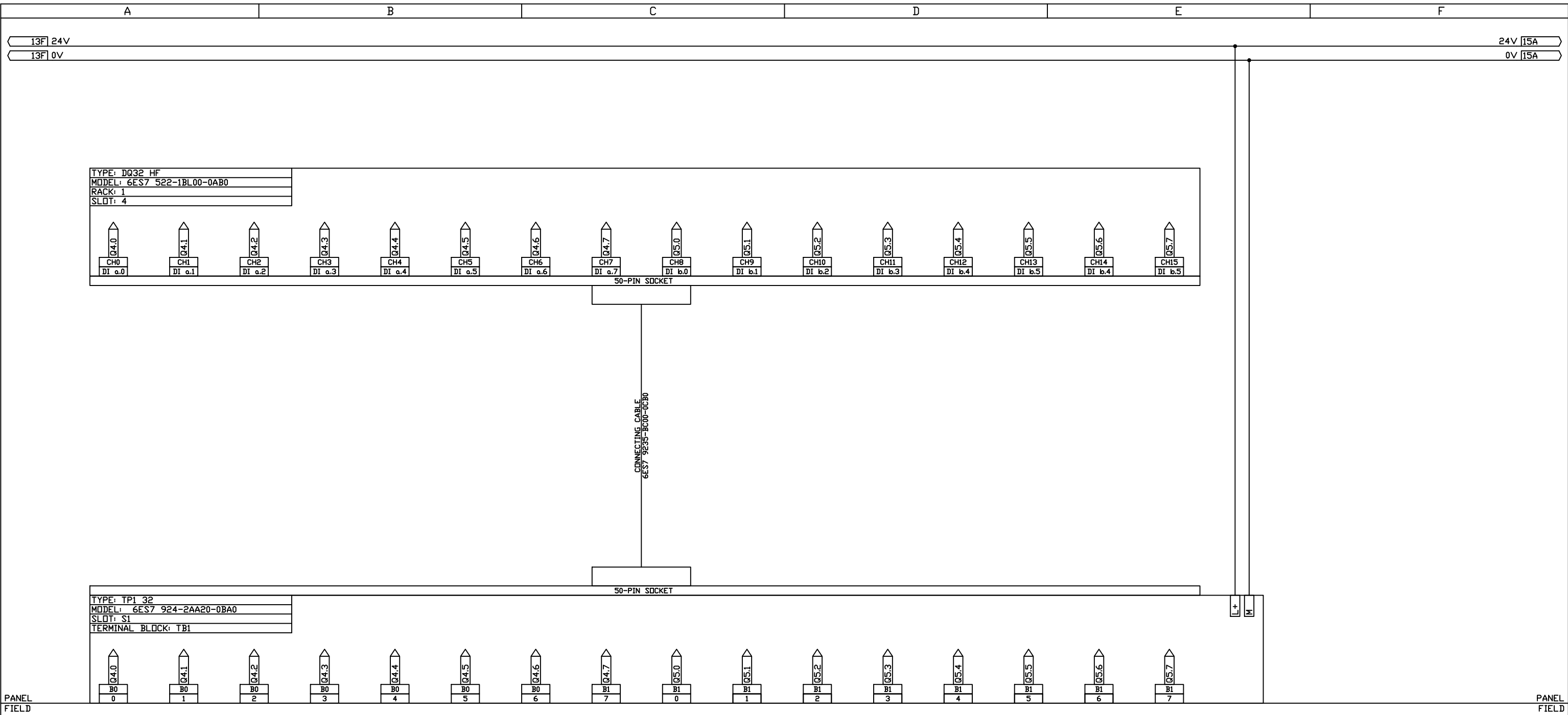
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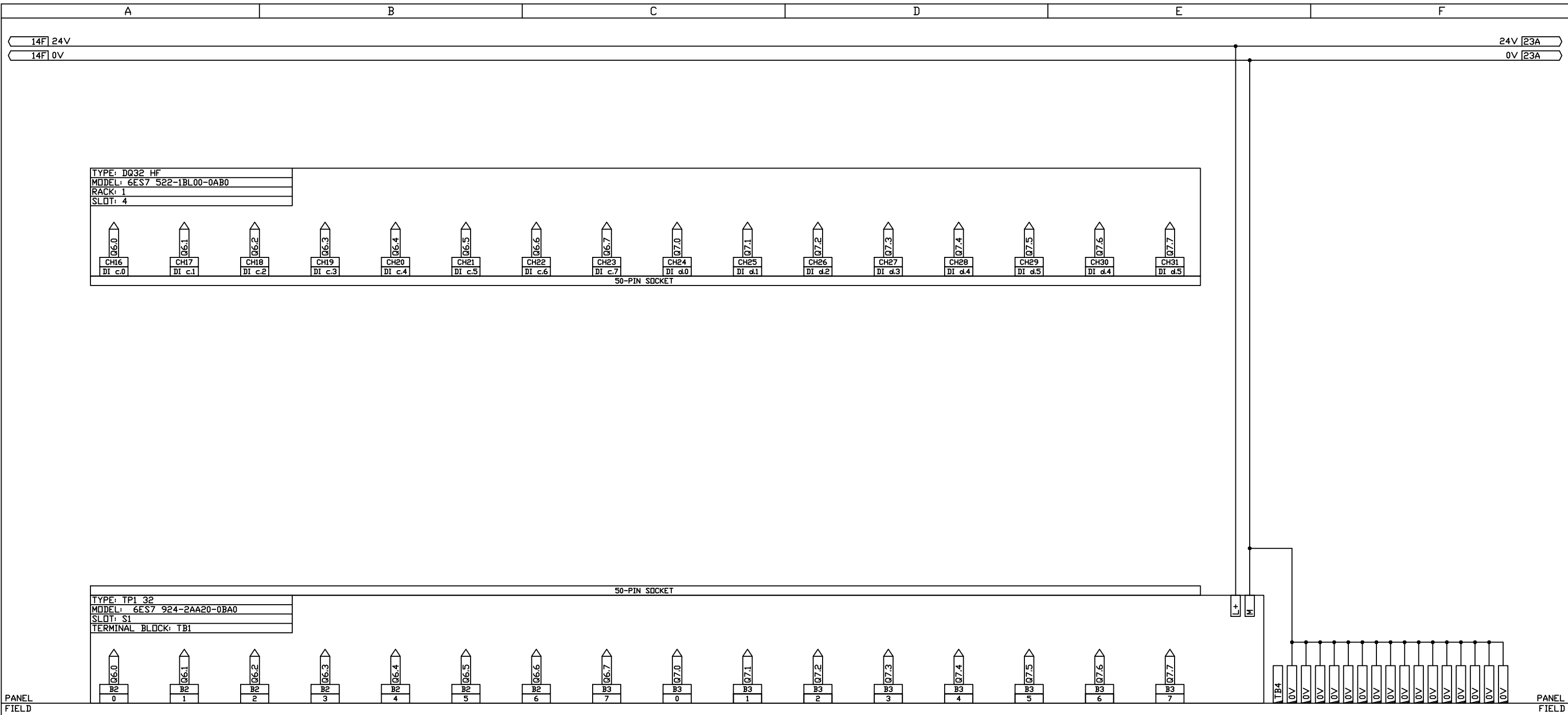




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PANEL FIELD

	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE			
HF HF Controls Ltd 100 Albert Drive, Glasgow, G41 2SJ Call 0800 783 9480 www.hf-group.co.uk Glasgow Edinburgh Manchester Belfast Aberdeen PLC RACK 1 SLOT 4 DIGITAL OUTPUT CARD		REV		DATE	DETAIL		CLIENT	PEEL PORTS		ENGINEERED		D.BALDOCK							
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TYPE: TP1 32

MODEL: 6ES7 924-2AA20-0BA0

SLOT: S1

TERMINAL BLOCK: TB1

Q6.0

B2

0

Q6.1

B2

1

Q6.2

B2

2

Q6.3

B2

3

Q6.4

B2

4

Q6.5

B2

5

Q6.6

B2

6

Q6.7

B3

7

Q7.0

B3

0

Q7.1

B3

1

Q7.2

B3

2

Q7.3

B3

3

Q7.4

B3

4

Q7.5

B3

5

Q7.6

B3

6

Q7.7

B3

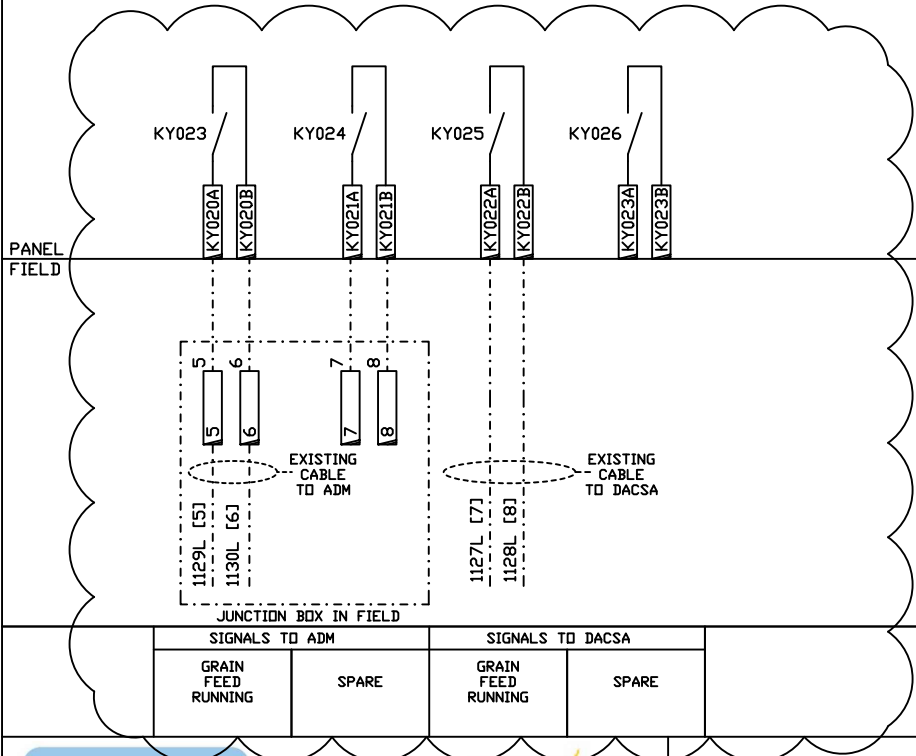
7

50-PIN SOCKET

PANEL FIELD

PANEL FIELD

	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	0VDC DISTRIBUTION TERMINALS																																						
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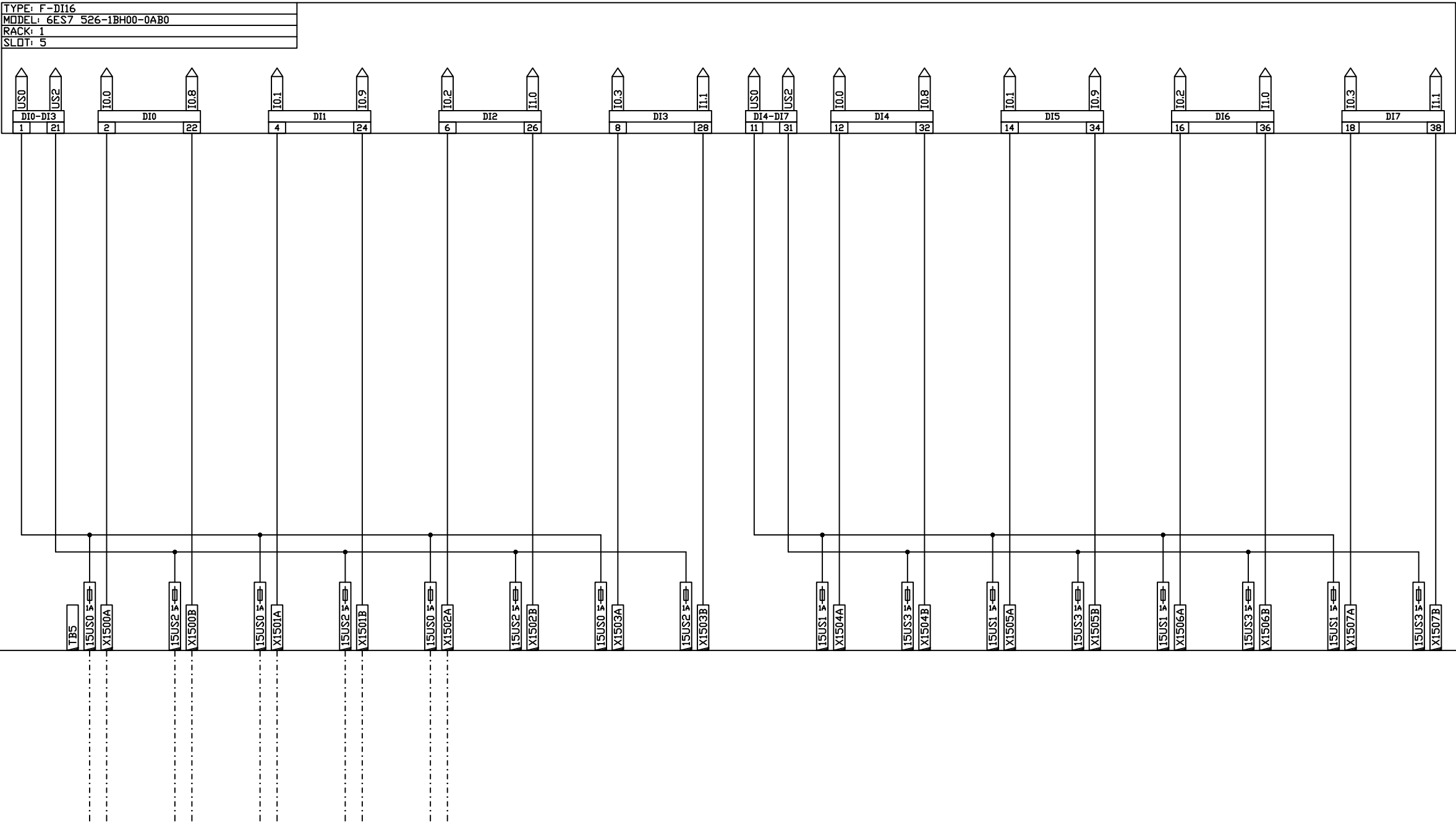
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